



Getting Started with Research Computing

From Command Line to Job Scheduling

Website: www.rc.colorado.edu

Documentation: curc.readthedocs.io

Support request form: colorado.service-now.com/req_portal?id=ucb_sc_rc_form



Meet the User Support Team



Layla
Freeborn



Brandon
Reyes



Andy
Monaghan



Michael
Schneider



John
Reiland



Dylan
Gottlieb

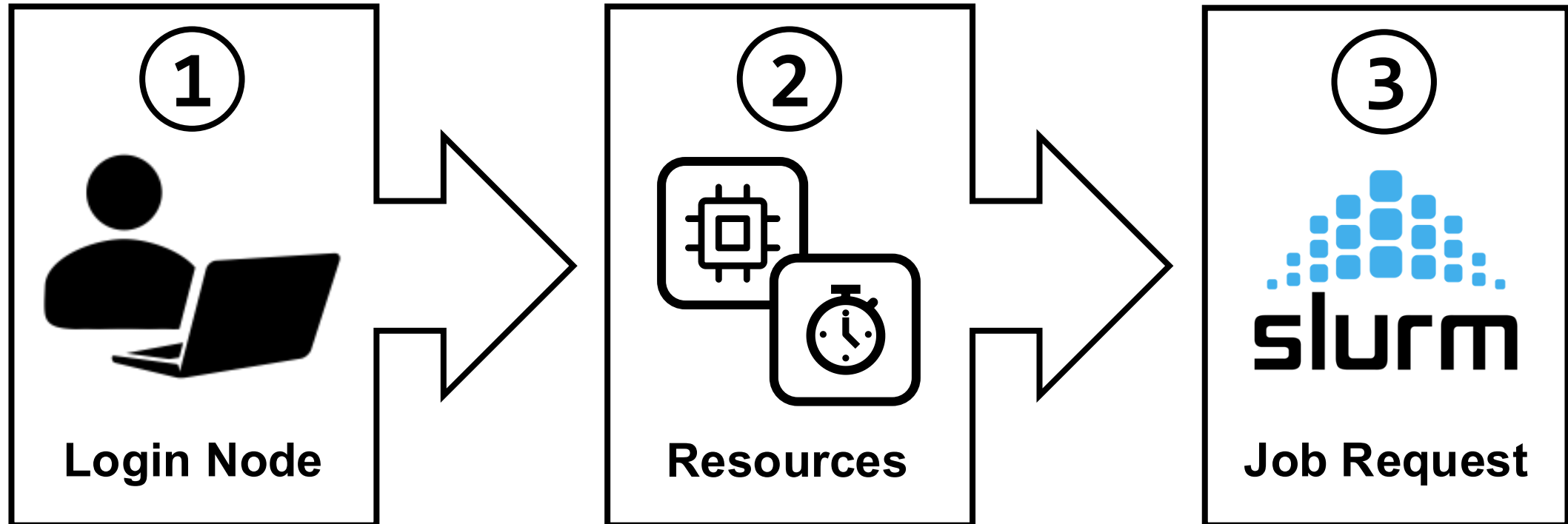


Mohal
Khandelwal



Ask Questions

General Overview of Job Submission



Login Nodes

A Login Node is the entryway onto our system. They are meant for:

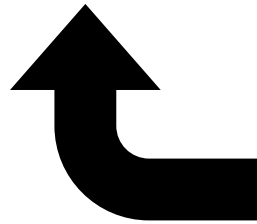
- Viewing or editing files
- Submitting Jobs

Login nodes should **never** be used to run code or software.

Once you are on a terminal, you will know you are on a login node by looking at the command prompt. If it contains “login”, you are on a login node

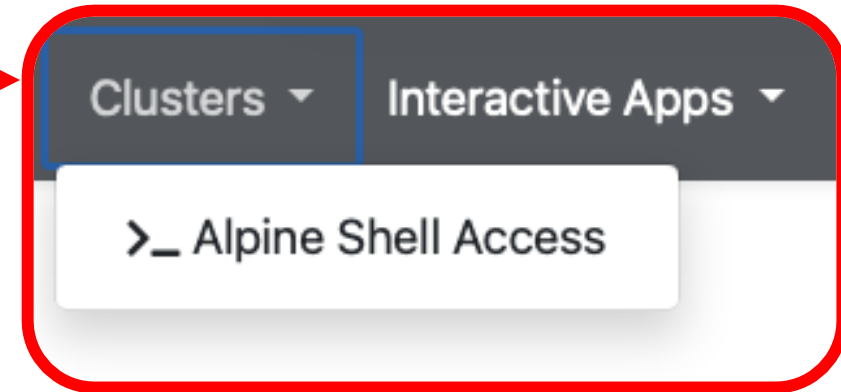
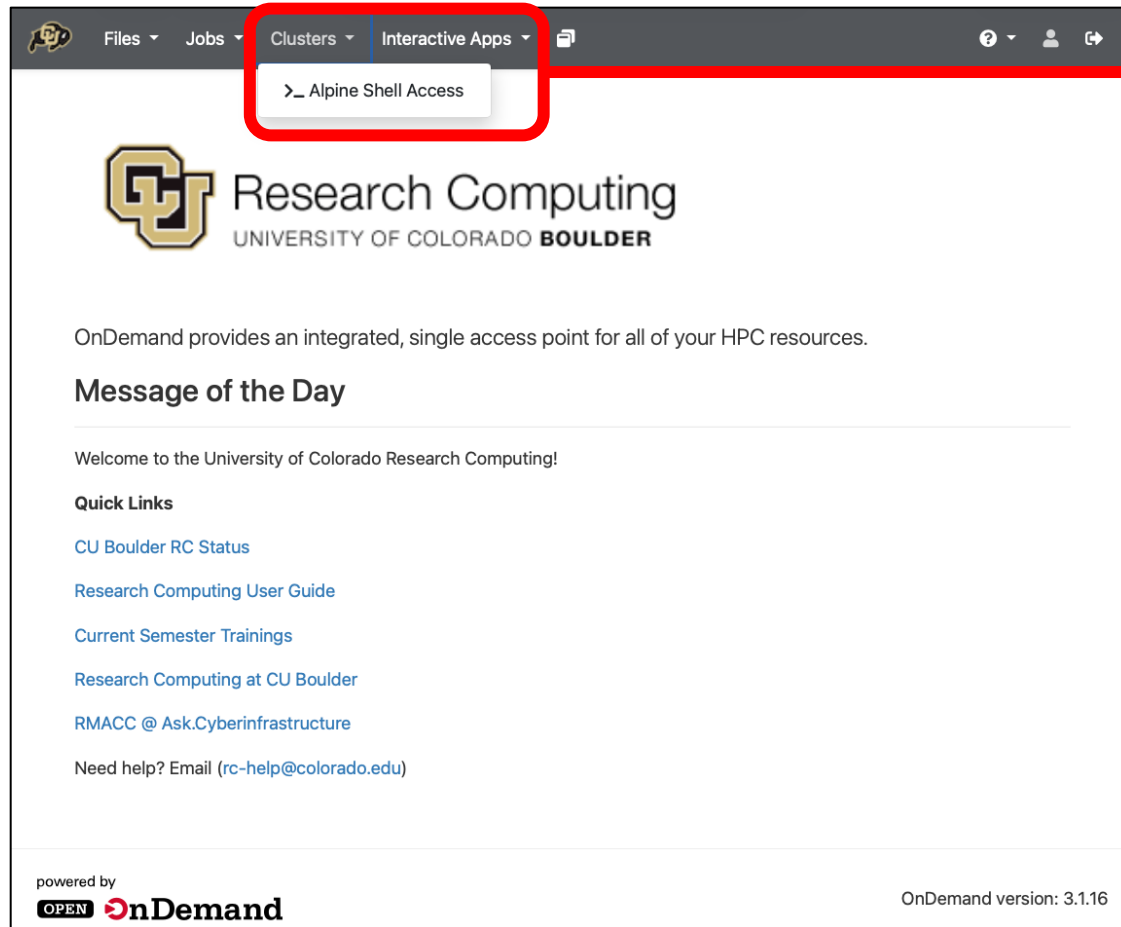
```
[ralphie@login-ci3]$
```

Accessing the Terminal



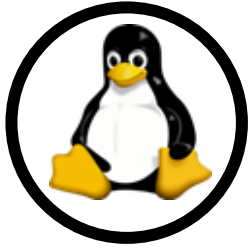
**Available to CU Boulder, CSU,
CU Anschutz**

Accessing the Terminal Cont.

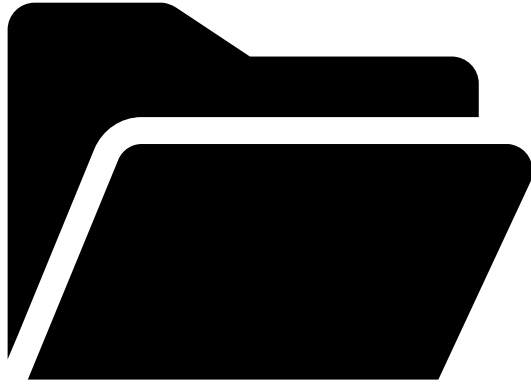


If you would like to access a terminal via ssh, please see:

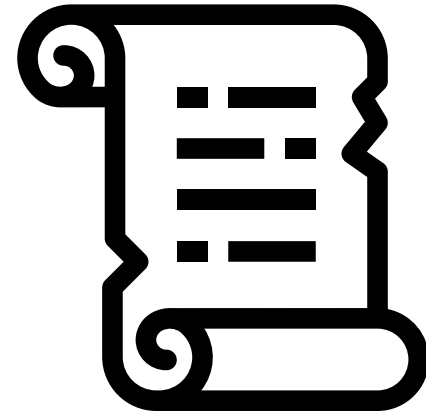
curc.readthedocs.io/en/latest/getting_started/logging-in.html#terminal-access-for-different-operating-systems



Linux Command Line



File Operations



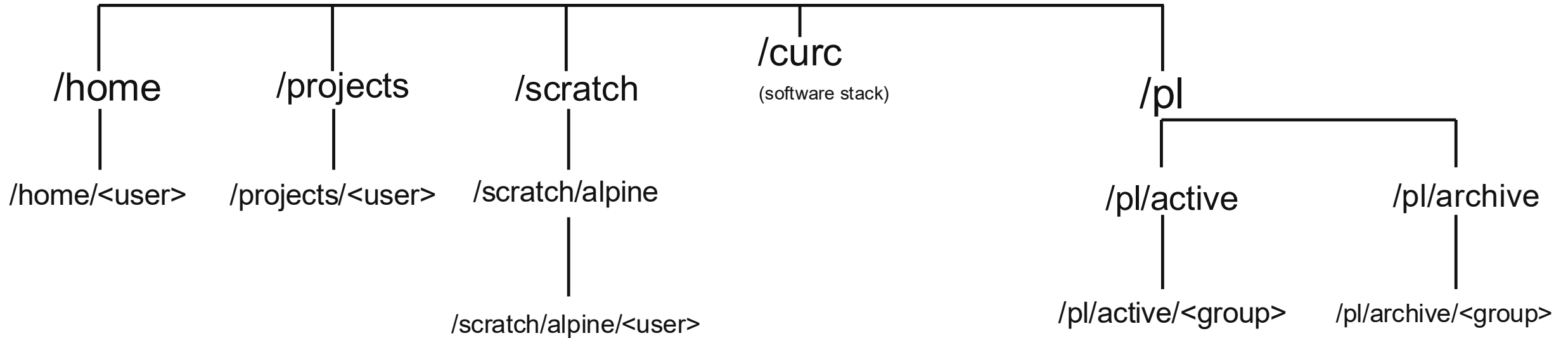
Scripting

List Files and Directories

`ls [option] [file/directory]`

- `-l` (long format)
- `-h` (readable file size)
- `-S` (sort by file size)
- `-t` (sort by modification time)
- `-a` (all files)
- `-r` (reverse sort)

Standard CURC Directories



File Operations

Change Directory

`cd [-o] <path>`

Make Directory

`mkdir [-o] <path>`

Copy / Move Files & Directories

`cp [-o] <source> <dest>`

Delete Files

`rm [-o] <path>`

Create and edit a file – Terminal Text Editors



Vim

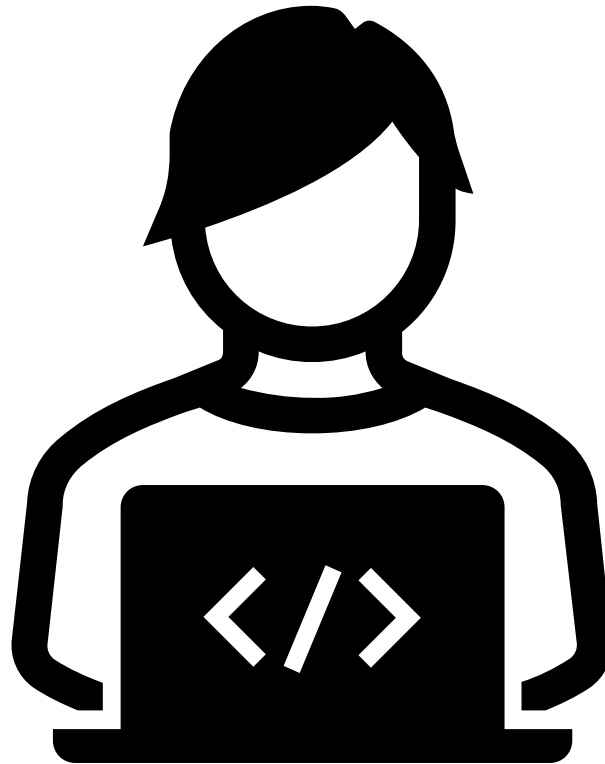
```
      :::  
      iLE88Dj.  :jD88888Dj:  
.LGitE888D.f8GjjjL8888E;  
iE  :8888Et.    .G8888.  
;i   E888,      ,8888,  
      D888,      :8888:  
      D888,      :8888:  
      D888,      :8888:  
      D888,      :8888:  
      888W,      :8888:  
      W88W,      :8888:  
      W88W:      :8888:  
      DGGD:      :8888:  
                      :8888:  
                      :W888:  
                      :8888:  
                      E888i  
                      tw88D
```

Nano



Emacs

Creating a Bash Script



Shell script basics

- In shell scripts, the first line must specify the shell e.g. `#!/bin/bash`
- The program loader recognizes the `#!` and will interpret the rest of the line (`/bin/bash`) as the interpreter program.
- If a line starts with `#`, it is a comment and is not run.

```
#!/bin/bash
```

```
#look at the files in /tmp
```

```
cd /tmp
```

```
ls
```

Shell to run

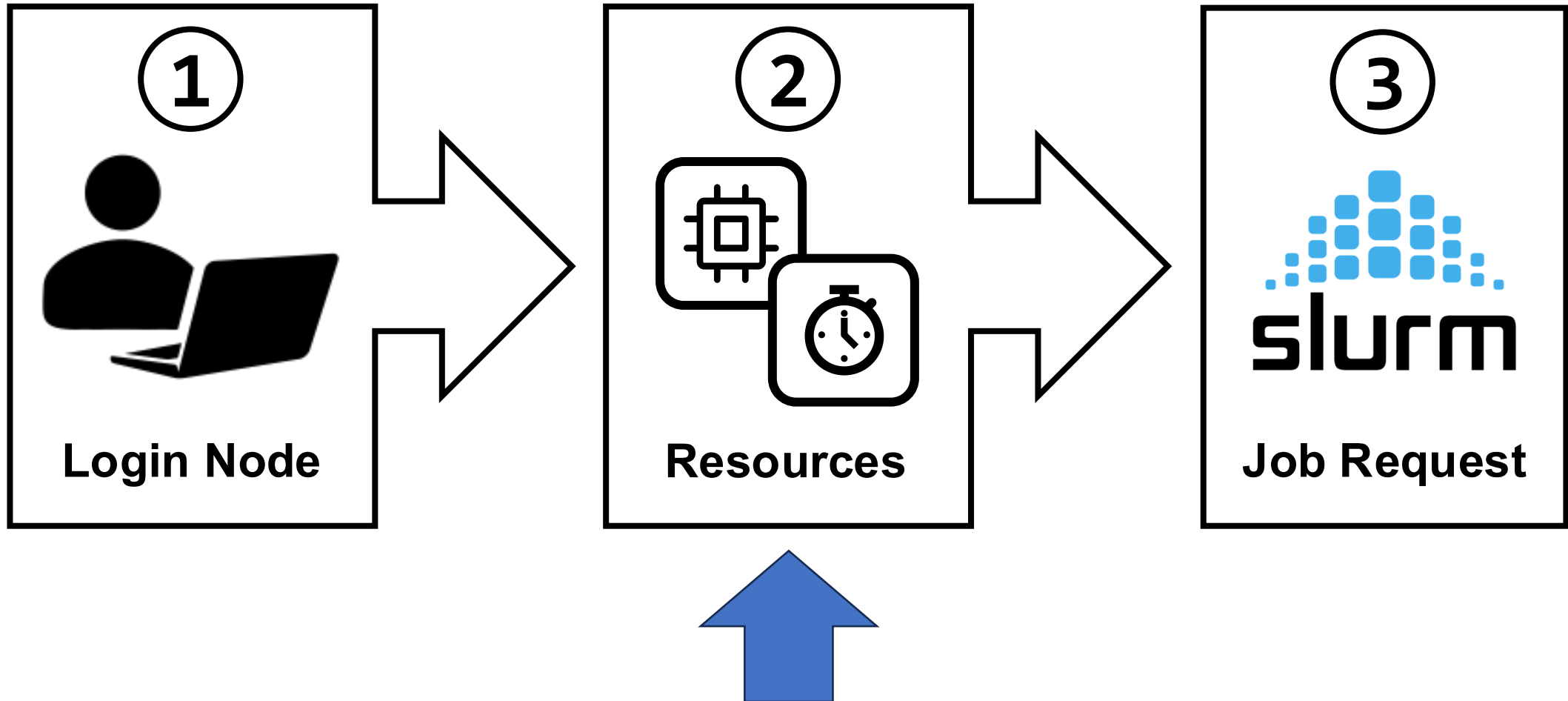
Comment

Change directories

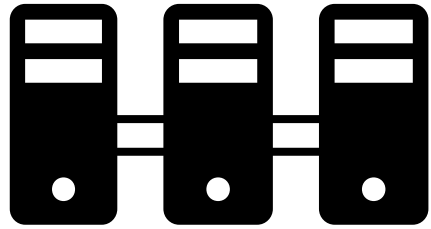
List everything in /tmp



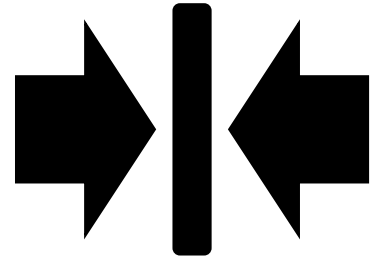
Selecting Resources



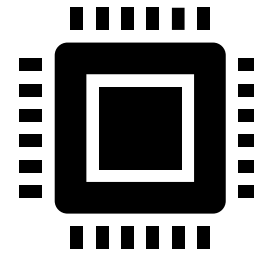
Common Resource Requests



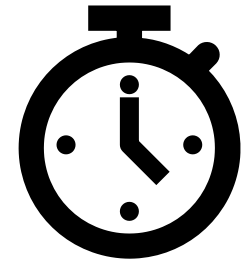
Partition



**Quality of
Service**



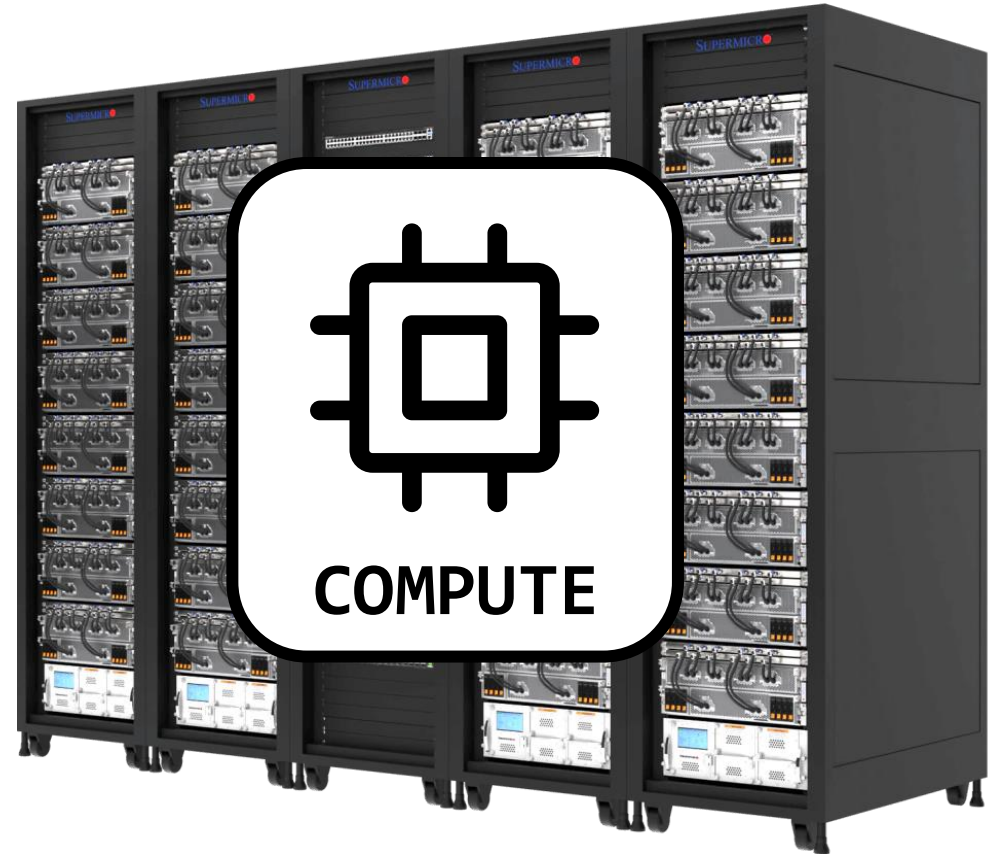
**Cores, GPUs,
and Nodes**



Wall Time

Alpine Partitions

- Partitions are a collection of compute nodes with common characteristics
- Compute nodes can be thought of as a desktop without a monitor
- For in-depth information on partitions see:
curc.readthedocs.io/en/latest/clusters/alpine/alpine-hardware.html#partitions



Alpine Partitions - acpu

acpu
General CPU



Alpine Partitions - amem

acpu

General CPU

amem

High Memory
CPU



Alpine Partitions – NVIDIA GPUs

acpu

General CPU

amem

High Memory
CPU



aa100

al40

ah200

artxpro6000

Nvidia GPUs



Alpine Partitions – AMD GPUs

acpu

General CPU

amem

High Memory
CPU



aa100

al40

ah200

artxpro6000

Nvidia GPUs

ami100

AMD GPUs

Alpine Testing Resources

- For several of the different types of hardware, we provide testing resources. These resources enable quicker access.
 - There are restrictions on the amount of resources you can request
- Access to testing resources is enabled through a combination of partition and QoS selection
- Available testing resources:
 - Standard CPUs
 - Compute nodes for verifying parallel applications
 - NVIDIA A100 and AMD MI100 GPUs

For more information see: curc.readthedocs.io/en/latest/clusters/alpine/alpine-hardware.html#special-purpose-resources

Quality of Service (QoS)

- Quality of Service specifies additional constraints for a given partition.
- On Alpine, QoS can be used to run long jobs, specify testing partitions, and select high-memory nodes

QoS	Description	Max wall time	Max jobs/user	Max nodes/user
normal	Default QoS	24 H	1000	128
long	For jobs needing longer wall times	7 D	200	20
mem	High-memory jobs	7 D	n/a	12
testing	For jobs submitted to testing partitions	1 H	1	n/a
compile	For jobs submitted to compile partition	12 H	1	n/a



Selecting Hardware

- Once a partition and QoS are selected, we need to say how much of the hardware we would like to use
- The most common selections are
 - The number of cores (often called “ntasks” on our system)
 - The number compute nodes (often referred to as “nodes”)
 - The type of GPU

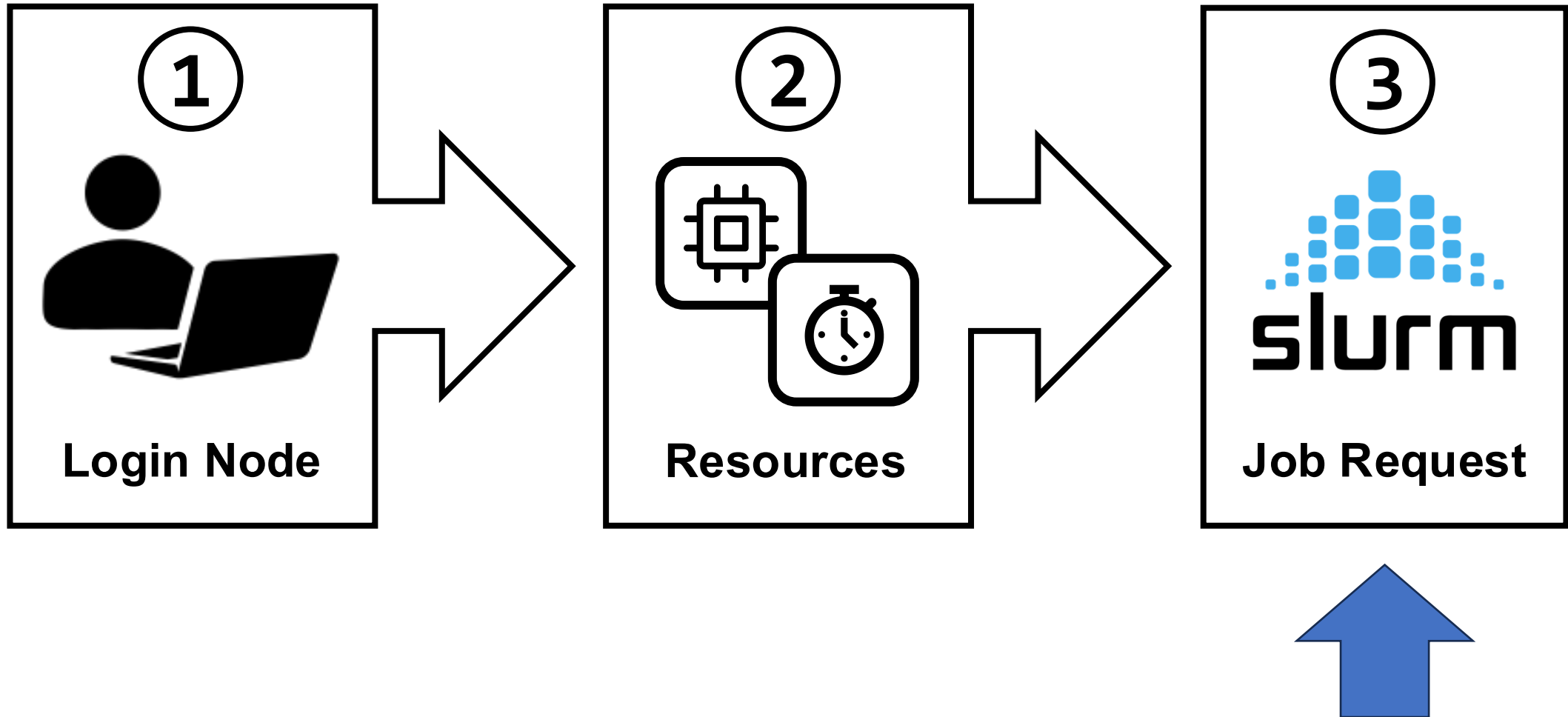
GRES Type

- Once
- MIG

Walltime

- Walltime or “run time” is how long you want to run on the selected resources
- How long do I specify?
 - Downsides to requesting too much time

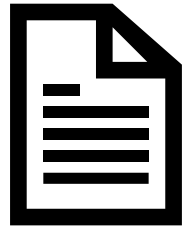
Requesting a Job



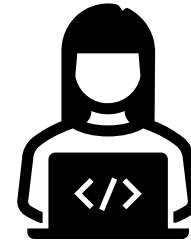
Planned Maintenance

Planned maintenance is ...

Batch vs Interactive Jobs



**Batch
Job**



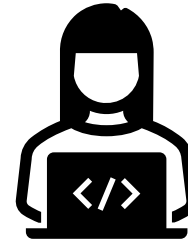
**Interactive
Job**

Batch Jobs

- Runs a job script
- No need for user input
- Best for tested workflows and long/big jobs



**Batch
Job**

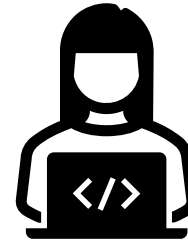


**Interactive
Job**

Interactive Jobs



**Batch
Job**



**Interactive
Job**

- Terminal or GUI
- Requires active user input
- Best for development and exploration

Slurm Directives

- Slurm is a popular resource manager for High-Performance Computing systems
 - Grabs the resources you are requesting
 - Determines when your job will run
- Slurm directives are how we tell Slurm what resources we would like
 - Are used for both interactive and batch jobs

Common Slurm Directives

- ★ • **Partition** – A collection of compute nodes
 - `--partition=<partition_name>`
 - ★ • **Quality of service (QoS)** – System defined constraints for a job (more on this later!)
 - `--qos=<qos>`
 - **Allocation** – Account to “charge to”
 - `--account=<account_name>`
 - **Number of nodes to run on**
 - `--nodes=<nodes>`
 - **Number of cores to run on**
 - `--ntasks=<number-of-tasks>`
- ★ - Must be specified for every job

Common Slurm Directives cont.

- ★ • **Wall time** – How long you want to run on these resources
 - `--time=<wall time>`
- **Job name**
 - `--job-name=<jobname>`
- **Output** – Where all output that would be written to the terminal should go
 - `--output=<name>`
- **Send an email when events happen in the job**
 - `--mail-type=<type>`
- **Email address to send updates to**
 - `--mail-user=<user>`

★ - Must be specified for every job

Starting an interactive job

acompile

- A custom CURC command
- Submits an interactive job to the acompile testing resources
- Limit of 4 CPUs for up to 12 hours
- Focused on compiling/building software and small workflows

sinteractive

- Request any partition, more resources, and longer walltime
- Subject to standard wait times (not immediate access)
- As the name implies, allows you to access a job interactively

Running an interactive job

- Here we will run a Python script using an interactive job
- First request resources:

```
[ralphie@login-ci3]$ sinteractive --partition=acpu --qos=cpu-normal --time=10 --ntasks=1 --nodes=1
```

- When the job starts you will be put on a compute node and you can execute your commands:

```
[ralphie@c3cpu-e2-u1]$ module load anaconda  
[ralphie@c3cpu-e2-u1]$ python my_very_cool_script.py
```

- To quit:

```
[ralphie@c3cpu-e2-u1]$ exit
```

Submitting a Batch Job

- **Batch Jobs** are jobs you submit to the scheduler that are run later without supervision
- A job script is simply a script that includes **SLURM directives** (resource specifics) ahead of any commands.

Anatomy of a job script

```
#!/bin/bash
```

```
## Directives
```

```
#SBATCH --<option>=<value>
```

```
## Software
```

```
module load <software>
```

```
## User scripting
```

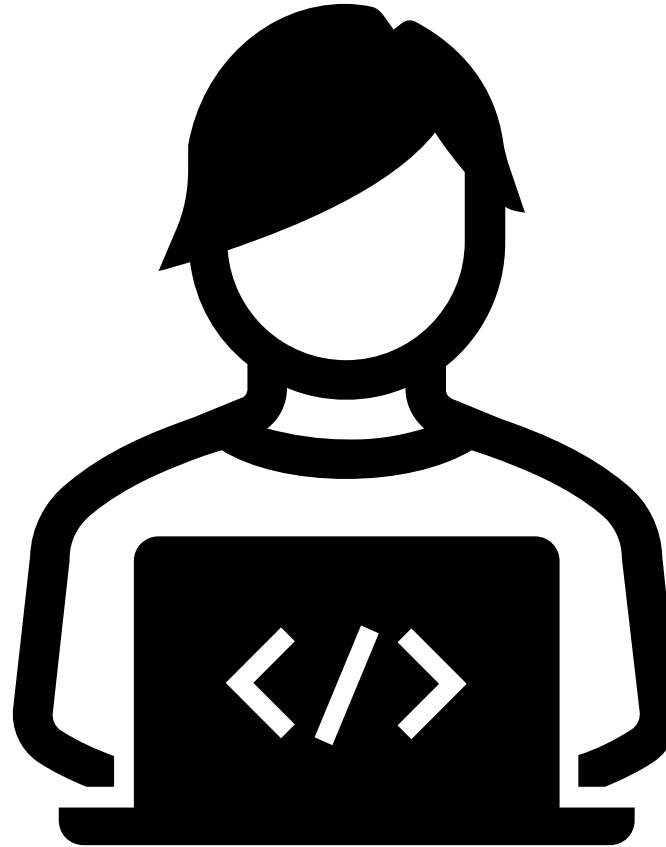
```
<command>
```



Directives in a job script

```
#SBATCH --<option>=<value>
```

Creating a Job Script



Example job script

```
#!/bin/bash

## Directives
#SBATCH --ntasks=1           # Number of requested tasks/cores
#SBATCH --time=0:01:00       # Max run time
#SBATCH --partition=acpu      # Specify Alpine CPU node
#SBATCH --qos=cpu-normal      # Specify QoS
#SBATCH --output=test_%j.out  # Rename standard output file

## Software
module purge                  # Purge all existing modules

## User commands
echo "This is a test submitted by $USER"
```



Submitting a Job script

`sbatch <job_file> <other-directives>`

- Command to submit a job to the Slurm job scheduler
- If we created the job script “my_first_job.sh” then we would submit it as follows:

`sbatch /path/to/my_first_job.sh`

- Modify Slurm directives in the batch script:

`sbatch /path/to/my_first_job.sh --ntasks=12`

Checking your jobs

- **squeue**: Monitor your jobs status **in queue and while running**:
 - By default, shows all jobs in queue can specify using:

```
$ squeue -u <username>  
$ squeue -p <partition>
```

- **sacct**: Check back on usage statistics of **previous Jobs**
 - By default, only checks all jobs from the start of the current day can specify using:

```
$ sacct -u <username>  
$ sacct --start=MM/DD/YY -u <username>  
$ sacct -j <job-id>
```

Checking your jobs

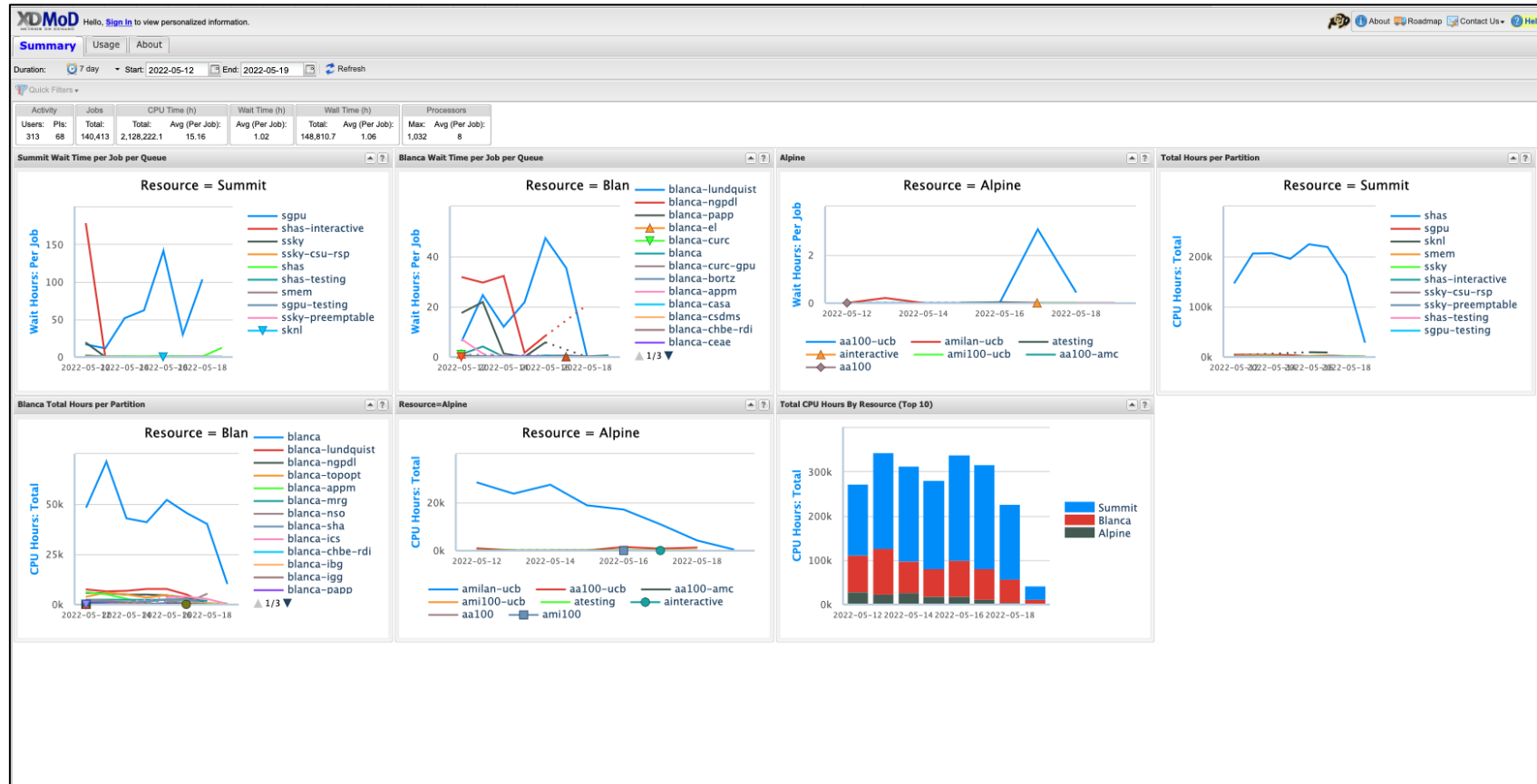
- To check the percentage of CPU and memory usage of a job **after it completes**, use `seff`

```
$ module load slurmttools  
$ seff <job number>
```

- To see general statistics for your submitted jobs, use `jobstats`

```
$ module load slurmttools  
$ jobstats <username> <# of days>
```

Job Monitoring - XDMOD



Help! I'm stuck, where do I go?

- **Documentation**: curc.readthedocs.io
- **Trainings with Center for Research Data and Digital Scholarship (CRDDS)**:
www.colorado.edu/crdds
- **Support request form**: colorado.service-now.com/req_portal?id=ucb_sc_rc_form
- **Consult Hours** (Tuesday 1:00-2:00pm in-person or virtually, Thursday 1:00-2:00pm virtually)

Survey and feedback

<https://s.zoom.us/m/bPITUoGhy>

